

Oliver Perrin

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SUMMARY

Machine Learning Engineer & Researcher with a dual focus on empirical research and production deployment. Shipped end-to-end ML pipelines at a pre-launch startup and authored research on multi-task learning in transformer architectures (EMNLP preprint). Builds deep learning systems from first principles, with a commitment to rigorous experimental methodology and ablation-driven analysis.

EXPERIENCE

Lead ML Engineer

Jan 2026 – Present

Stealth Startup

Boone, NC

- Sole MLE on a 10-person pre-launch team (founder + 8 SWEs); transitioned from senior-year intern to full-time post-graduation, owning all ML architecture, research, and production decisions in direct collaboration with the founder ahead of launch.
- Shipped production attribute and unit harmonization system: SBERT embeddings over 125K+ product attributes, cosine-similarity retrieval, and GPU-accelerated batch inference, with confidence metadata and top-k match evidence surfaced for output explainability.
- Designed schema-agnostic ingestion that auto-detects multiple JSON formats and normalizes them into a canonical attribute representation; built unit-harmonization logic across physical, electrical, and thermal dimensions, with output now consumed by the company's database, browser extension, mobile app, and web platform.
- Built a real-time ML operations dashboard (FastAPI + WebSockets) with batch upload, step-level event streaming, and grouped downloadable outputs for debugging and stakeholder demos.
- Prototyping a second system — a multi-agent harmonization and product-data scraping pipeline built with Claude Agent Skills orchestrated through Obsidian — and presenting architecture deep-dives to the company's advisory board (*specifics under NDA*).

Systems Engineering Intern

June 2023 – August 2024

Triaxis Power Consulting LLC

Charlotte, NC

- Scoped, specified, and deployed an IoT monitoring system (Raspberry Pi 4 with thermocouples and potentiometers) to track pipe temperature and thermal expansion in fossil fuel power plants for predictive maintenance; owned hardware selection, setup, and data visualization end-to-end.
- Researched, specified, and configured NAS hardware and a NAS-to-SharePoint sync workflow to centralize 200GB+ of engineering data across projects, replacing ad-hoc manual transfers.

PROJECTS

LexiMind: Multi-Task Learning for Literary & Academic Text | *PyTorch, Transformers*

Nov 2025 – Present

- Implemented a 272M-parameter encoder-decoder transformer from scratch (Pre-LN, RMSNorm, T5 relative position bias, FlashAttention, gated-GELU FFN, KV-cache) with explicit FLAN-T5-base weight mapping, then jointly trained on abstractive summarization, topic classification, and multi-label emotion detection across literary and academic domains.
- Diagnosed negative transfer in naive MTL and designed two targeted interventions — learned attention pooling for classification heads and temperature-based task sampling ($\alpha=0.5$) — improving emotion-detection sample-averaged F1 from 0.199 $\rightarrow$ 0.352 (+77%) and surpassing the single-task baseline; per-class threshold tuning further raised macro F1 to 0.290 on held-out test.
- Conducted full ablation across pooling strategies, task scheduling, and initialization, plus encoder-only BERT-base baselines to isolate architecture vs. MTL effects; shared-encoder MTL outperforms the BERT topic-classification specialist (85.7% vs. 83.1% accuracy).
- Targeting arXiv preprint and EMNLP 2026 submission; 5-seed validation run underway, with bootstrap CIs, cross-task MD5 deduplication, gradient-conflict diagnostics, and MLflow experiment tracking already in place.

EDUCATION

Appalachian State University

Boone, NC

B.S. Computer Science, Minor in Mathematics, Data Science Certificate

May 2026

- Relevant Coursework: Linear Algebra, Statistical Data Analysis, Numerical Methods, Computational Mathematics, Advanced Reinforcement Learning, Theoretical Computer Science, Applied Machine Learning, Data Structures & Algorithms
- Research Interests: Multi-task learning, reinforcement learning theory, NLP / language modeling
- Senior Capstone (ML Lead): Built ML harmonization system and real-time inference dashboard; deliverables deployed in production at partnering stealth startup and carried forward into current full-time role.
- Notable Projects: Formal language interpreter with lambda calculus and type inference (Haskell), pipelined CPU simulator (C/C++), 2D game engine (Java)

TECHNICAL SKILLS

Machine Learning: PyTorch, Transformers (Hugging Face), TensorFlow, Scikit-learn, Multi-task Learning, Transfer Learning, Representation Learning, NLP, Reinforcement Learning

Data & Analysis: Pandas, NumPy, Plotly, Seaborn, SQL, Statistical Modeling, MLflow, Weights & Biases

Engineering & Systems: Python, C/C++, Java, JavaScript, FastAPI, React.js, PostgreSQL, Docker, CI/CD, L^AT_EX